

Nonlinear autopilot control design for a 2-DOF helicopter model

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Abstract: The paper illustrates a robust control scheme for application to helicopters in vertical flight mode (both take-off and landing) to guarantee altitude stabilisation. A nonlinear helicopter model is used to derive the proposed control in which Lyapunov's direct method is used to establish the overall system stability. A recursive design technique is applied to design a nonlinear robust controller using the highly coupled system structure. It will be shown that the proposed robust controller provides semiglobal stabilisation of uniform ultimate boundedness for achieving the desired altitude. That is, the control design is valid for all values of the helicopter's collective pitch angle away from zero. The vertical flight application demonstrates a unique robust control mechanisation for helicopter and V/STOL autopilot augmentation systems.

1 Introduction

Because of multivariable nonlinear dynamics of helicopter systems, most of the existing results to date have been based on the linearisation model or through several linearisation techniques [1, 2]. These linearised results only provide local stability and are liable to be unstable with unstructured perturbations and uncertainties. In this paper, robust control schemes seem to be very attractive since they compensate for both the lumped nonlinearities of the helicopter system and its parameter uncertainties. Furthermore, the robust control design developed in this manuscript guarantees semiglobal uniform ultimate boundedness stability in comparison with linear controllers that yield a small stabilisation region around an operation point. Two types of robust stabilisation methods of a helicopter model have been shown in [2]. It was shown that the Lyapunov-based (quadratic) robust

stabilisation had a better overall performance than the H^∞ robust stabilisation. Actually, Lyapunov-based design is commonly used for stabilisation of nonlinear systems where classical techniques yield poor results.

In this paper, under hovering conditions as well as take-off and landing, we develop a nonlinear robust controller for the altitude control based on Lyapunov's direct method [3, 4] using the nonlinear model directly – that is, without the need for any linearisation of the model or the use of linearisation techniques for the system transformation. Using the highly coupled system structure, the decomposition of system dynamics can be established. Then, the recursive design technique (backstepping) developed in [5, 6] is used to design the robust control in which a fictitious control is designed for each decomposed system and subsystem, and finally the fictitious controllers are embedded into the overall control strategy. Since the robust controller is designed for the nonlinear model using Lyapunov stability criteria, semiglobal stabilisation of the system is guaranteed. The proposed scheme also provides smooth hovering along with take-off and landing since it is robust against disturbances resulting from uncertainties due to unmodelled system states.

This paper is organised as follows. Section 2 formulates the helicopter model and the robust control design procedure. The simulation results for the proposed nonlinear model are presented in Section 3. Concluding remarks are given in Section 4.

2 Problem formulation and robust control design

Consider the nonlinear helicopter model given by [7]:

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= a_0 + a_1 x_2 + a_2 x_2^2 + (a_3 + a_4 x_4 - \sqrt{a_5 + a_6 x_4}) x_3^2 \\ \dot{x}_3 &= a_7 + a_8 x_3 + [a_9 \sin(x_4) + a_{10}] x_3^2 + u_1 \\ \dot{x}_4 &= x_5 \\ \dot{x}_5 &= a_{11} + a_{12} x_4 + a_{13} x_3^2 \sin x_4 + a_{14} x_5 + u_2 \\ x &= [h \quad \dot{h} \quad \omega \quad \theta \quad \dot{\theta}]^T \\ u &= [u_1 \quad u_2]^T = [K_1 u_{th} - K_2 u_\theta]^T\end{aligned}\quad (1)$$

where

h = height above ground (metres)

ω = rotational speed of the rotor blades (rad/s)

θ = collective pitch angle of rotor blades (rad)

u_{th} = input to the throttle

u_θ = input to collective servomechanisms.

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The nominal values in [7] for constants K_i and a_i are given to be: $K_1 = -0.1088 \text{ s}^{-2}$, $K_2 = 0.25397 \text{ s}^{-2}$, $a_0 = -17.67 \text{ m/s}^2$, $a_1 = a_2 = -0.1 \text{ s}^{-1}$, $a_3 = 5.31 \times 10^{-4}$, $a_4 = 1.5364 \times 10^{-2}$, $a_5 = 2.82 \times 10^{-7}$, $a_6 = 1.632 \times 10^{-5}$, $a_7 = -13.92 \text{ s}^{-2}$, $a_8 = -0.7 \text{ s}^{-1}$, $a_9 = a_{10} = -0.0028$, $a_{11} = 434.88 \text{ s}^{-2}$, $a_{12} = -800 \text{ s}^{-2}$, $a_{13} = -0.1$ and $a_{14} = -65 \text{ s}^{-1}$.

Due to system perturbation, one can assume without loss of generality that nominal values a_i and K_i could fluctuate with an amount $\hat{a}_i$ and $\hat{K}_i$, respectively, i.e. $a_i = \bar{a}_i \pm \hat{a}_i$ and $K_i = \bar{K}_i \pm \hat{K}_i$. The perturbations in a_i and K_i are bounded by:

$$|\hat{a}_i| < \bar{a}_i \quad i = 0, 1, 2, 7, \dots, 14$$

$$|\hat{K}_i| < \bar{K}_i \quad i = 1, 2$$

To demonstrate that eqn. 1 satisfy the generalised matching conditions [6], i.e. the system looks more like a chain of integrators, a nonlinear state transformation is proposed as follows:

$$\zeta_1 = x_1$$

$$\zeta_2 = x_2$$

$$\zeta_3 = x_3^2$$

$$\zeta_4 = x_4$$

$$\zeta_5 = x_5$$

Through this transformation, eqn. 1 become

$$\dot{\zeta}_1 = \zeta_2$$

$$\dot{\zeta}_2 = a_0 + a_1 \zeta_2 + a_2 \zeta_2^2 + \Upsilon \zeta_3^2$$

$$\dot{\zeta}_3 = 2a_7 \zeta_3^2 + 2a_8 \zeta_3 + 2[a_9 \sin \zeta_4 + a_{10}] \zeta_3^3 + 2\zeta_3^2 u_1$$

$$\dot{\zeta}_4 = \zeta_5$$

$$\dot{\zeta}_5 = a_{11} + a_{12} \zeta_4 + a_{13} \zeta_3 \sin \zeta_4 + a_{14} \zeta_5 + u_2 \quad (2)$$

where $\Upsilon = a_3 a_4 \zeta_4 - \sqrt{(a_5 + a_6 \zeta_4)}$.

Remark: Since the rotor blade speed $\omega = x_3$ is never zero while the helicopter is in flight, the design of control u_1 will yield no singularity problem.

Since the above system satisfies the generalised matching conditions, a robust controller can be designed following the design procedure outlined in [6]. The first step in designing a robust controller for a system constructed via a chain of integrators is to utilise the backstepping technique. This is accomplished by introducing fictitious controls for ζ_1 , ζ_2 , ζ_3 , ζ_4 and ζ_5 denoted by ζ_1^d , ζ_2^d , ζ_3^d , ζ_4^d and ζ_5^d , respectively. Then, dynamic equations of eqn. 2 in terms of transformed state variables $z_i = \zeta_i - \zeta_i^d$, $i = 1, 2, 3, 4, 5$, become:

$$\dot{z}_1 = z_2 - \dot{\zeta}_1^d + \dot{\zeta}_2^d$$

$$\dot{z}_2 = a_0 + a_1(z_2 + \zeta_2^d) + a_2(z_2 + \zeta_2^d)^2 + \Upsilon z_3 - \dot{\zeta}_2^d + \Upsilon \zeta_3^d$$

$$\begin{aligned} \dot{z}_3 = & 2a_7(z_3 + \zeta_3^d)^2 + 2a_8(z_3 + \zeta_3^d) \\ & + 2[a_9 \sin(z_4 + \zeta_4^d) + a_{10}](z_3 + \zeta_3^d)^3 \\ & + 2(z_3 + \zeta_3^d)^2 u_1 - \dot{\zeta}_3^d \end{aligned}$$

$$\dot{z}_4 = z_5 - \dot{\zeta}_4^d + \dot{\zeta}_5^d$$

$$\begin{aligned} \dot{z}_5 = & a_{11} + a_{12}(z_4 + \zeta_4^d) + a_{13}(z_3 + \zeta_3^d) \sin(z_4 + \zeta_4^d) \\ & + a_{14}(z_5 + \zeta_5^d) - \dot{\zeta}_5^d + u_2 \end{aligned} \quad (3)$$

The importance of introducing the state transformation that yields fictitious controllers can be demonstrated as follows. The main goal behind state transformation $z_i = \zeta_i^d - \zeta_i$ is to design robust controllers u_1 and u_2 that guarantee semiglobal uniform ultimate boundedness for eqn. 3. In other words, the aim is to drive states z_i

to zero or very close to zero within arbitrary small error, $z_i \rightarrow 0$. Consequently, state ζ_i is converging to a desired state ζ_i^d ($\zeta_i \rightarrow \zeta_i^d$). As a result, one can choose a time-varying desired trajectory (or simply a constant) for the height that the helicopter will track. For instance, one can simply choose a height value, say $\zeta_1^d = 10\text{m}$, or a time-varying desired trajectory, say $\zeta_1^d = 10 \sin tm$ (an up and down motion). This state transformation allows more general choices for desired trajectories, i.e. one can choose a set-point tracking or a trajectory tracking.

Remark: Note that there exists a singularity in fictitious control ζ_3^d for eqn. 3 in the null set where $\Upsilon = 0$. This occurs when the collective pitch angle is zero, i.e. $x_4 = \zeta_4 = 0$. However, the collective pitch angle is usually nonzero in a practical situation (typically 7 degrees). For all purposes, the following control design is valid for all values of pitch angle away from zero. We shall name this region the 'semiglobal' region. Now, let us proceed with the backstepping design procedure. The robust control design procedure is broken into five main steps, as follows:

1. In the first subsystem of eqn. 3, since ζ_1^d is known, fictitious control ζ_2^d can be chosen to be

$$\zeta_2^d = -kz_1 + \dot{\zeta}_1^d \quad (4)$$

where gain $k > 0$ is a constant chosen by the designer.

Let sub-Lyapunov function $V_1(z_1)$ for first subsystem of eqn. 3 be $V_1(z_1) = 1/2 z_1^2$; then, differentiating $V_1(z_1)$ along the trajectory of z_1 yields

$$\dot{V}_1 = z_1 \dot{z}_1$$

$$= z_1 z_2 - z_1 \dot{\zeta}_1^d + z_1 (-kz_1 + \dot{\zeta}_1^d)$$

$$\leq -k_1^2 + \|z_1\| \|z_2\| \quad (5)$$

2. The second subsystem of eqn. 3 can be rewritten as

$$\dot{z}_2 = a_0 + a_1 \zeta_2 + a_2 \zeta_2^2 - \dot{\zeta}_2^d + \Upsilon z_3 + \Upsilon \zeta_3^d \pm \text{sgn}(z_2) z_1 \quad (6)$$

where the term $\text{sgn}(z_2) z_1$ is used to cancel term $\|z_1\| \|z_2\|$ found in eqn. 5.

The total matched uncertainty in eqn. 6 can be bounded as

$$\begin{aligned} & \|\hat{a}_0 + \hat{a}_1 \zeta_2 + \hat{a}_2 \zeta_2^2 + \text{sgn}(z_2) z_1 - \dot{\zeta}_2^d\| \\ & \leq \|\hat{a}_0\| + \|\hat{a}_1\| \|\zeta_2\| + \|\hat{a}_2\| \|\zeta_2^2\| + \|z_1\| + \|\dot{\zeta}_2^d\| \\ & \leq \bar{a}_0 + \bar{a}_1 \|\zeta_2\| + \bar{a}_2 \|\zeta_2^2\| + \|z_1\| + k \|\dot{z}_1\| + \|\dot{\zeta}_1^d\| \\ & \leq \rho_2(z_1, z_2) \end{aligned}$$

Let sub-Lyapunov function $V_2(z_2)$ for eqn. 6 be of the form $V_2(z_2) = 1/2 z_2^2$; then differentiating $V_2(z_2)$ along the trajectory of z_2 yields

$$\dot{V}_2 = (a_0 + a_1 \zeta_2 + a_2 \zeta_2^2 + \text{sgn}(z_2) z_1 - \dot{\zeta}_2^d) z_2$$

$$+ \Upsilon z_2 z_3 + \Upsilon z_2 \zeta_3^d - \|z_2\| z_1$$

$$\leq \|z_2\| \rho_2(z_1, z_2) + \|\Upsilon\| \|z_2\| \|z_3\| + \Upsilon z_2 \zeta_3^d$$

$$- \|z_2\| \|z_1\| + [a_0 + a_1 \zeta_2 + a_2 \zeta_2^2] z_2$$

Select fictitious control ζ_3^d to be of the form

$$\begin{aligned} \zeta_3^d(z_1, z_2) = & -\Upsilon^{-1} \left[k z_2 + a_1 + a_1 \zeta_2 + a_2 \zeta_2^2 \right. \\ & \left. + \frac{\mu_2 \|\mu_2\|^2}{\|\mu_2\|^3 + \epsilon_2^3} \rho_2(z_1, z_2) \right] \end{aligned}$$

where $\epsilon_2 > 0$ is a constant chosen by the designer and

$$\mu_2 = z_1 \rho_2(z_1, z_2)$$

Substituting ζ_3^d in the above inequality yields

$$\begin{aligned}\dot{V}_2 &\leq -k_z^2 + \|\mu_2\| + \|\Upsilon\| \|z_2\| \|z_3\| \\ &\quad - z_2 \left[\frac{\mu_2 \|\mu_2\|^2}{\|\mu_2\|^3 + \epsilon_3^3} \rho_2(z_1, z_2) \right] - \|z_1\| \|z_2\| \\ &\leq -k_z^2 + \|\Upsilon\| \|z_2\| \|z_3\| + \epsilon_2 - \|z_1\| \|z_2\| \quad (7)\end{aligned}$$

Now, let us first combine eqns. 5 and 7; we get

$$\dot{V}_1 + \dot{V}_2 \leq -k(z_1^2 + z_2^2) + \|\Upsilon\| \|z_2\| \|z_3\| + \epsilon_2 \quad (8)$$

3. Next, we need to transform the third subsystem of eqn. 3 via state transformation $z_3 = \zeta_3 - \zeta_3^d$, which yields

$$\begin{aligned}\dot{z}_3 &= 2a_8\zeta_3 + [2a_7 + 2(a_9 \sin \zeta_4 + a_{10})\zeta_3]x_3 \\ &\quad - \dot{\zeta}_3^d + 2x_3u_1 \pm \Upsilon z_2 z_3\end{aligned} \quad (9)$$

where the last term in eqn. 9 is used to cancel term $\|\Upsilon\| \|z_2\| \|z_3\|$ in eqn. 8. Furthermore, one can bound total uncertainties in eqn. 9 as

$$\begin{aligned}&\|2\hat{a}_8\zeta_3 + 2\hat{a}_7x_3 + 2(\hat{a}_9 \sin \zeta_4 + \hat{a}_{10})x_3\zeta_3 \\ &\quad - \dot{\zeta}_3^d + \Upsilon z_2 z_3\| \\ &\leq 2\bar{a}_9\|\zeta_3\| + 2\bar{a}_7\|x_3\| + 4\bar{a}_9\|x_3\| \|\zeta_3\| + \|\dot{\zeta}_3^d\| \\ &\quad + \|\Upsilon\| \|z_2\| \|z_3\| \\ &\leq \rho_3(z_1, z_2, z_3)\end{aligned}$$

where

$$\begin{aligned}\|\dot{\zeta}_3^d\| &\leq \left\| \frac{\partial \zeta_3^d}{\partial z_1} \right\| (\|z_2\| + 2\rho_1 + k\|z_1\|) \\ &\quad + \left\| \frac{\partial \zeta_3^d}{\partial z_2} \right\| (|a_0| + |a_1| \|\zeta_2\| + \|\Upsilon\| \|x_3\| + \|\dot{\zeta}_2^d\|)\end{aligned}$$

To choose control u_1 , let sub-Lyapunov function $V_3(z_3)$ for subsystem eqn. 9 be of the form $V_3(z_3) = \frac{1}{2}z_3^2$; then, taking the derivative of $V_3(\cdot)$ along the trajectory of z_3 , we obtain

$$\begin{aligned}\dot{V}_3 &\leq \|z_3\| \rho_3 + 2x_3z_3u_1 - \|\Upsilon\| \|z_2\| \|z_3\| + 2a_8z_3\zeta_3 \\ &\quad + [2a_7 + 2(a_9 \sin \zeta_4 + a_{10})\zeta_3]x_3z_3\end{aligned} \quad (10)$$

Select throttle control u_1 to be of the form

$$\begin{aligned}u_1(z_1, z_2, z_3, z_4) &= -\frac{1}{2}x_3^{-1} \left\{ k z_3 + 2a_8\zeta_3 \right. \\ &\quad + [2a_7 + 2(a_9 \sin \zeta_4 + a_{10})\zeta_3]x_3 \\ &\quad \left. + \frac{\mu_3}{\|\mu_3\| + \epsilon_3} \rho_3(z_1, z_2, z_3, z_4) \right\}\end{aligned} \quad (11)$$

where $\epsilon_3 > 0$ is a constant chosen by the designer and $\mu_3 = z_3\rho_3(z_1, z_2, z_3, z_4)$. Substituting u_1 in inequality eqn. 10 yields

$$\begin{aligned}\dot{V}_3 &\leq -kz_3^2 + \|\mu_3\| - \frac{\mu_3^2}{\|\mu_3\| + \epsilon_3} - \|\Upsilon\| \|z_2\| \|z_3\| \\ &\leq -kz_3^2 + \epsilon_3 - \|\Upsilon\| \|z_2\| \|z_3\|\end{aligned} \quad (12)$$

Combining eqns. 8 and 12, we obtain

$$\dot{V}_1 + \dot{V}_2 + \dot{V}_3 \leq -k(z_1^2 + z_2^2 + z_3^2) + \epsilon_2 + \epsilon_3 \quad (13)$$

As a result, one can conclude that for the choice of throttle control given by eqn. 11, states z_1 , z_2 and z_3 are uniformly ultimately bounded within arbitrary small proximity around the origin.

4. Next, we continue with the design of the collective servomechanism control. In the fourth subsystem of

eqn. 3, since ζ_4 denotes the collective pitch angle of the rotor blades and since ζ_4^d is known, then fictitious control ζ_5^d can be chosen to be

$$\zeta_5^d = -kz_4 + \dot{\zeta}_4^d \quad (14)$$

Let sub-Lyapunov function $V_4(z_4)$ for the fourth subsystem of eqn. 3 be $V_4(z_4) = \frac{1}{2}z_4^2$; then, differentiating $V_4(z_4)$ along the trajectory of z_4 yields

$$\dot{V}_4 = z_4(z_5 - \dot{\zeta}_4^d + \dot{\zeta}_5^d) \leq -kz_4^2 + \|z_4\| \|z_5\| \quad (15)$$

5. The final step is to design u_2 to dominate all terms in the last subsystem of eqn. 3. The last subsystem of eqn. 3 can be rewritten as

$$\begin{aligned}\dot{z}_5 &= a_{11} + a_{12}\zeta_4 + a_{13}\zeta_3 \sin \zeta_4 + a_{14}\zeta_5 \\ &\quad - \dot{\zeta}_5^d + u_2 \pm \text{sgn}(z_5)z_4\end{aligned}$$

where the last term in the above equation is used to cancel term $\|z_4\| \|z_5\|$ in eqn. 15. Let the following terms be bounded:

$$\begin{aligned}&\|\hat{a}_{11} + \hat{a}_{12}\zeta_4 + \hat{a}_{13} \sin \zeta_4 + \hat{a}_{14}\zeta_5 - \dot{\zeta}_5^d + \text{sgn}(z_5)z_4\| \\ &\leq \bar{a}_{11} + \bar{a}_{12}\|\zeta_4\| + \bar{a}_{13}\|\zeta_3\| + \bar{a}_{14}\|\zeta_5\| + \|z_4\| + \|\dot{\zeta}_5^d\| \\ &\leq \rho_5(z_1, z_2, z_3, z_4, z_5)\end{aligned}$$

where

$$\|\dot{\zeta}_5^d\| \leq k(\|x_5\| + \|\zeta_4^d\|) + \ddot{\zeta}_4^d$$

To design control u_2 , choose sub-Lyapunov function $V_5(z_5) = \frac{1}{2}z_5^2$; then, taking the derivative of $V_5(\cdot)$ along the trajectory of z_5 yields

$$\begin{aligned}\dot{V}_5 &\leq \|z_5\| \rho_5 - \|z_4\| \|z_5\| + z_5u_2 \\ &\quad + (a_{11} + a_{12}\zeta_4 + a_{13}\zeta_3 \sin \zeta_4 + a_{14}\zeta_5)z_5\end{aligned} \quad (16)$$

Select control u_2 to be of the form

$$\begin{aligned}u_2(z_1, z_2, z_3, z_4, z_5) &= -kz_5 - a_{11} - a_{12}\zeta_4 - a_{13}\zeta_3 \sin \zeta_4 - a_{14}\zeta_5 \\ &\quad - \frac{\mu_5}{\|\mu_5\| + \epsilon_5} \rho_5(z_1, z_2, z_3, z_4, z_5)\end{aligned} \quad (17)$$

where $\epsilon_5 > 0$ is a constant chosen by the designer and

$$\mu_5 = z_5\rho_5(z_1, z_2, z_3, z_4, z_5)$$

Substituting u_2 in inequality eqn. 16 yields

$$\begin{aligned}\dot{V}_5 &\leq -kz_5^2 + \|\mu_5\| - \frac{\mu_5^2}{\|\mu_5\| + \epsilon_5} - \|z_4\| \|z_5\| \\ &\leq -kz_5^2 + \epsilon_5 - \|z_4\| \|z_5\|\end{aligned} \quad (18)$$

Combining eqns. 15 and 18, we obtain

$$\dot{V}_4 + \dot{V}_5 \leq -k(z_4^2 + z_5^2) + \epsilon_5 \quad (19)$$

As a result, one can conclude that for choice of control u_2 given by eqn. 17, states z_4 and z_5 are uniformly ultimately bounded within arbitrary small proximity around the origin.

To conclude the semiglobal stability of eqn. 3, combine eqns. 13 and 19 to yield

$$\begin{aligned}\dot{V}_1 + \dot{V}_2 + \dot{V}_3 + \dot{V}_4 + \dot{V}_5 &\leq -k(z_1^2 + z_2^2 + z_3^2 + z_4^2 + z_5^2) + \epsilon_2 + \epsilon_3 + \epsilon_5\end{aligned}$$

or

$$\dot{V} \leq -2kV + \epsilon'$$

where $V = \sum V_i$ for $i = 1, \dots, 5$, and $\epsilon' = \sum \epsilon_i$ for $i = 2, 3, 5$. Solving for the above differential equation yields

$$\|V\|_\infty \leq \frac{\epsilon'}{2k}$$

and as a result one can conclude semiglobal uniform ultimate boundedness stability for eqn. 3 with a stability region given by

$$\|z_i\|_\infty \leq \sqrt{\frac{\epsilon'}{k}}$$

which can be made arbitrarily small by choosing a large gain k . Since our interest is in eqn. 1 and more precisely output state x_1 , one can easily demonstrate that state x_1 tracks a desired trajectory with an error no larger than

$$\|x_1^d - x_1\|_\infty \leq \sqrt{\frac{\epsilon'}{k}}$$

3 Simulation

The need for highly reliable and stable (in the presence of gusts) 'zero-zero' landing and take-off capability for helicopters and V/STOL class vehicles has increased morbidly for critical passenger or cargo delivery to intercity locations and/or to other places in close proximity to population centres. The robust control scheme just described should be strongly considered for this application. Embedding this concept into the autopilot augmentation systems of helicopter and/or V/STOL vehicles and coupling with a highly accurate navigation system (inertial, GPS, radar, etc.) could provide such a 'zero-zero' approach and landing system. Since the altitude state is the most critical for hovering and landing and take-off manoeuvres, the control mechanisation of the vertical trajectory profile will be demonstrated.

To illustrate the performance of the nonlinear robust control design procedure for the nonlinear helicopter system given by eqn. 1, we present two simulation examples: the first one addresses the take-off and landing trajectory, while the second example presents the hovering case. We simulate both examples using SIMNON®.

Example 1: Consider the helicopter model in eqn. 1 with the following choices:

1. The desired vertical flight is $x_1^d = \sin(0.5t)$ m, desired pitch angle $x_4^d = 0.05$ rad, gain $k = 5$ and $\epsilon_i = 1$ for $i = 2, 3, 5$.

2. Initial conditions are chosen to be:

$$\begin{aligned} x_1 &= 0 \text{ m} & x_2 &= 0 \text{ m/s} \\ x_3 &= 200 \text{ rad/s} & x_4 &= 0.15 \text{ rad} \\ x_5 &= 0 \text{ rad/s} \end{aligned}$$

3. Unstructured perturbations in a_i and K_i are chosen with 10% tolerance, i.e. $\hat{a}_i = 0.1a_i$ and $\hat{K}_i = 0.1K_i$.

4. Uncertainties are bounded by the following bounding functions:

$$\begin{aligned} \rho_2 &= 1.767 + 0.01\|\zeta_2\| + 0.01\|\zeta_2\|^2 + (k^2 + 1)\|z_1\| \\ &\quad + k\|z_2\| + 2k\|\dot{\zeta}_1^d\| + \|\ddot{\zeta}_1^d\| \\ \rho_3 &= 0.14\|\zeta_3\| + 2.784\|x_3\| + 0.00123\|x_3\|\|\zeta_3\| \\ &\quad + 0.1\|z_2\|\|z_3\| + \|\dot{\zeta}_3^d\| \\ \rho_5 &= 43.5 + 80\|\zeta_4\| + 0.01\|\zeta_3\| + 6.5\|\zeta_5\| \\ &\quad + \|z_4\| + k\|x_5\| + k\|\zeta_4^d\| \end{aligned}$$

The simulation results are given in Figs. 1 and 2 for the state variables and controls, respectively. One can notice from these Figures that the helicopter model tracks a given desired trajectory with accurate proximity.

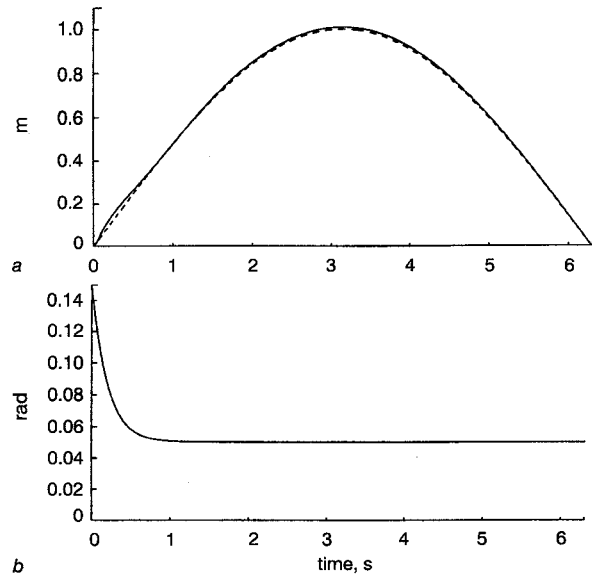


Fig. 1 Plots of states x_1 and x_4 for a take-off and landing trajectory
--- desired — calculated
a Trajectory tracking b Pitch angle

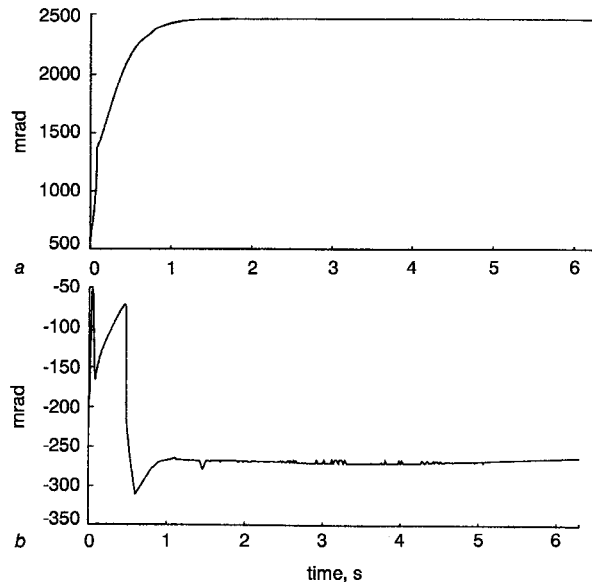


Fig. 2 Plots of controls u_1 and u_2 for a take-off and landing trajectory
a Throttle control b Servomechanical control

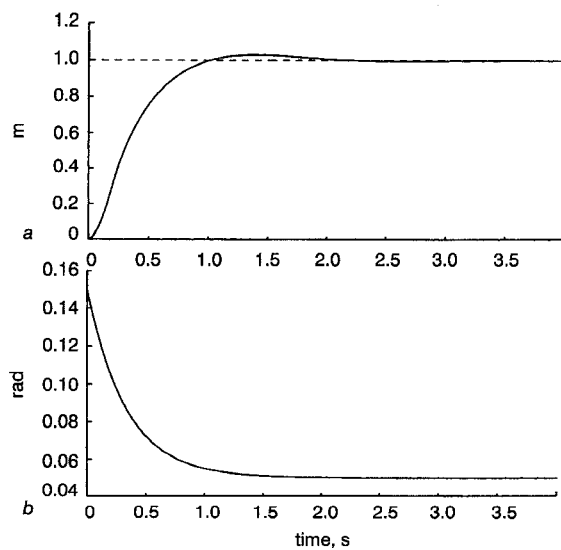


Fig. 3 Plots of states x_1 and x_4 for a hovering trajectory
--- desired — calculated
a Set-point tracking b Pitch angle

Example 2: In this example, we present the case where the helicopter is hovering. We simulate eqn. 1 with the same choices given in Example 1, except that $x_1^d = 1\text{m}$ and $k = 3$. The simulation results are given in Figs. 3 and 4 for the state variables and controls, respectively.

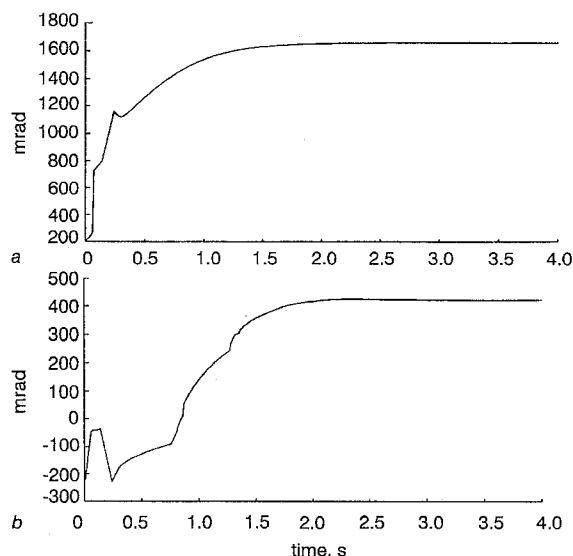


Fig. 4 Plots of controls u_1 and u_2 for a hovering trajectory
a Throttle control b Servomechanical control

4 Conclusion

In this paper we have demonstrated a nonlinear robust controller for a helicopter in vertical flight. The control

design was based on Lyapunov's direct method without the need for system linearisation. Moreover, we showed that this control design scheme is robust against perturbation in the system parameters. The control design for the helicopter model presented here is based on a step-by-step design scheme that would eventually lead to the robust controller we are interested in. The next step would be to apply this design technique to the 4 DOF (degree of freedom) helicopter model and eventually to the more complicated 6 DOF model. The advantage of designing a nonlinear controller is to allow a helicopter system to operate outside its limited linear region. This is usually preferred for helicopters with high and complex attack manoeuvres.

5 References

- 1 GARRAD, W., and LOW, E.: 'Eigenspace design of helicopter flight control systems'. Technical report, Dept. of Aerospace Engineering and Mechanics, Univ. of Minnesota, Nov. 1990
- 2 KIENITZ, K., WU, Q., and MANSOUR, M.: 'Robust stabilization of a helicopter model'. Proceedings of the 29th CDC, 1990, pp. 2607-2612
- 3 KHALIL, H.K.: 'Nonlinear systems' (Macmillan, New York, 1992)
- 4 SLOTINE, J., and LI, W.: 'Applied Nonlinear Control' (Prentice-Hall, Englewood Cliffs, NJ, 1991)
- 5 KANELAKOPOULOS, I., KOKOTOVIC, P.V., and MORSE, A.S.: 'Systematic design of adaptive controllers for feedback linearizable systems', *IEEE Trans. Autom. Control*, 1991, **36**, pp. 1241-1253
- 6 QU, Z.: 'Robust control of nonlinear uncertain systems under generalized matching conditions', *Automatica*, 1993, **29**, pp. 985-998
- 7 RAMIREZ, H.S., ZRIBI, M., and AHMAD, S.: 'Dynamical sliding mode control approach for vertical flight regulation in helicopters', *IEE*, January 1994, pp. 19-24